

# TASK-2

To measure distance using HC-SR04 sensor and an ESP32.

## The MicroPython Code

```
import machine
import time

# Define GPIO pins based on your setup
TRIG_PIN = 15
ECHO_PIN = 16

# Initialize pins
# Trig needs to be an OUTput to send the pulse
trig = machine.Pin(TRIG_PIN, machine.Pin.OUT)
# Echo needs to be an INput to listen for the bounce back
echo = machine.Pin(ECHO_PIN, machine.Pin.IN)

def get_distance():
    # Ensure trigger is low initially
    trig.value(0)
    time.sleep_us(5)

    # Send a 10-microsecond pulse to trigger the sensor
    trig.value(1)
    time.sleep_us(10)
    trig.value(0)

    # Measure the duration of the incoming pulse on the Echo pin (in microseconds)
    # 30000 microseconds is a timeout (~5 meters max range) to prevent freezing
    duration = machine.time_pulse_us(echo, 1, 30000)

    if duration < 0:
        # If the reading timed out or failed
        return None

    # Calculate distance in centimeters:
    # Speed of sound is ~0.0343 cm/us.
    # We divide by 2 because the sound travels to the object and back.
    distance = (duration * 0.0343) / 2
    return distance

# Main loop
```

```
print("Starting distance measurement...")
while True:
    dist = get_distance()

    if dist is not None:
        print(f"Distance: {dist:.2f} cm")
    else:
        print("Out of range or sensor error")

    # Wait half a second before the next reading
    time.sleep(0.5)
```